

Q. check whether the given ring $(\mathbb{Z}_8, +, \cdot)$ is zero div or not?

→ Q. $\mathbb{Z}_8 = \{ \bar{0}, \bar{1}, \bar{2}, \bar{3}, \bar{4}, \bar{5}, \bar{6}, \bar{7} \}$

$\mathbb{Z}_8$

$\bar{2} \cdot \bar{4} = \bar{0}$

$\bar{4} \cdot \bar{6} = \bar{0}$

$\bar{4} \cdot \bar{4} = \bar{0}$

zero divisors of $\mathbb{Z}_8 = \{ \bar{2}, \bar{4}, \bar{6} \}$

$(\mathbb{Z}_8, +, \cdot)$ is a ring with zero divisors.

Q. HW → $(\mathbb{Z}_{10}, +, \cdot)$.

* Definitⁿ for Integral Domain :-

A ring commutative ring with unity & without zero divisors is called an integral domain.

$(\mathbb{R}, +, \cdot)$ is a ring with unity

$$a * e = e * a = a$$

$$a * e = a$$

$$a \cdot e = a$$

$$e = a/a = 1$$

$$\boxed{e=1}$$

$$1 \in \mathbb{R}$$

ii) $(\mathbb{R}, +, \cdot)$ is commutative ring.

$$a * b = a \cdot b$$

$$= b \cdot a$$

$$= b + a$$

iii) $(\mathbb{R}, +, \cdot)$ is a ring without zero divisor.

Since, there does not exist $(\exists)$
 $a \neq 0, b \neq 0$, in $\mathbb{R}$ such that
 $a \cdot b = 0$

$\Rightarrow (\mathbb{R}, +, \cdot)$ is an integral domain. $\triangle$

check $(\mathbb{Z}, +, \cdot)$ is an I.D.

(i) $(\mathbb{Z}, +, \cdot)$ is a ring with unity.

$$a * e = e * a = a$$

$\therefore$ we take $a * e = a$.

$$e = a/a = 1. \quad e = 1 \in \mathbb{R}$$

(ii) $(\mathbb{Z}, +, \cdot)$ is commutative ring

$$a * b = a \cdot b \\ = b \cdot a = b * a$$

(iii) $(\mathbb{Z}, +, \cdot)$ is a ring without zero divisors.

Since, $\nexists a \neq 0, b \neq 0$ in $\mathbb{Z}$ such that $a \cdot b = 0$

$\Rightarrow (\mathbb{Z}, +, \cdot)$ is an integral domain.

$\Rightarrow$ H.W. $(\mathbb{Q}, +, \cdot)$ is an ID.

$\Rightarrow$ $\mathbb{Q}$. check whether $(\mathbb{Z}/6, +_6, \cdot_6)$ is an ID.

$\Rightarrow$ (1) $(\mathbb{Z}/6, +_6, \cdot_6)$ with unity

$$a * e = e * a = a$$

$$a \cdot_6 e = a$$

$$e = 1 \in \mathbb{Z}/6$$

(ii) $(\mathbb{Z}/6, +_6, \cdot_6)$ is commutative ring

$$a * b = a \cdot b \\ = b \cdot a \\ = b * a$$

iii) $(\mathbb{Z}_6, +, \cdot)$ is a ring without zero divisor?

$$\overline{2} \cdot \overline{3} = \overline{0} \text{ in } \mathbb{Z}_6 \\ \text{but } \overline{1} \cdot \overline{6} = \overline{0}$$

$\Rightarrow (\mathbb{Z}_6, +, \cdot)$ is not integral domain

NOTE $\Rightarrow (\mathbb{Z}/n, +, \cdot)$ is an ID only if $n = \text{prime}$.

Field :-

A commutative ring with unity is called field, if every non-zero element has multiplicative inverse.

Q. Is $(\mathbb{R}, +, \cdot)$ is a field?

$\Rightarrow$ (i) $(\mathbb{R}, +, \cdot)$ is a ring with unity.

$$a * e = e * a = a$$

$$\therefore a * e = a$$

$$a \cdot e = a$$

$$e = a/a = 1$$

$$1 \in \mathbb{R}$$

(ii) $(\mathbb{R}, +, \cdot)$ is commutative ring
 $a * b = a \cdot b = b \cdot a = b * a$
 $\forall a, b \in \mathbb{R}$

(iii) Let $a \neq 0$ in $\mathbb{R}$,
 $a^{-1} = b$
 $a * b = b * a = e$
 $a * b = e$
 $a \cdot b = 1$
 $b = 1/a \in \mathbb{R}$

every $a \neq 0$, in $\mathbb{R}$ has multiplicative inverse.

$(\mathbb{R}, +, \cdot)$ is a field.

H.W. Q. $(\mathbb{Q}, +, \cdot)$ is a field?

Q. $(\mathbb{Z}, +, \cdot)$ is a field?

⇒ (i) $(\mathbb{Z}, +, \cdot)$ is a ring with unity.

$$\begin{aligned} a * e &= e * a = a \\ a * e &= a \\ a \cdot e &= a \\ e &= a/a = 1 \\ 1 &\in \mathbb{R} \end{aligned}$$

(ii) $(\mathbb{Z}, +, \cdot)$ is commutative ring.

$$\begin{aligned} a * b &= a \cdot b = b \cdot a = b * a \\ \forall a, b &\in \mathbb{R} \end{aligned}$$

(iii) Let $a \neq 0$ in $\mathbb{R}$,
 $a^{-1} = b$
 $a * b = b * a = e$
 $a * b = e$
 $a \cdot b = 1$
 $b = \frac{1}{a}$

$(\mathbb{R}, +, \cdot)$ is ~~a~~ not a field.

★ Every field is an integral domain
 but converse need not be true.
 Then it is not field.

Proof:-

Let $(\mathbb{F}, +, \cdot)$ is a field.

$\Rightarrow \mathbb{F}$ is a ring, a commutative ring
 with unity & every non-zero
 element has multiplicative inverse.

To prove that $\mathbb{F}$ is an integral
 domain to prove $\mathbb{F}$ is without zero
 divisor

Let $a \neq 0$,

$\Rightarrow a^{-1}$ exist in IF

$$a^{-1} (a \cdot b) = a^{-1} \cdot 0$$

$$(a^{-1} \cdot a) \cdot b = 0 \dots \text{by associativity}$$

$$1 \cdot b = 0$$

$$\therefore b = 0$$

$(IF, +, \cdot)$ is an ID.

* Every finite I.D is a field.

* Subring:-

Let R be ring & let S be a non-empty subset of R then S is subring of R if foll. conditions holds:-

$$\begin{aligned} 1) & a - b \in S & \forall a, b \in S \\ 2) & a \cdot b \in S & \forall a, b \in S \\ & \quad \vee a + (-b) \end{aligned}$$

Q Show that $S = \{0, 3\}$ is a subring of $(\mathbb{Z}_6, +, \cdot)$

$\rightarrow$ The composition Table of S under $+ \& \cdot$

$+$	0	3
0	0	3
3	3	0

$\cdot$	0	3
0	0	0
3	0	3

from composition table for $a, b \in S$
 $a, b \in S \& a \cdot b \in S$ Hence $S = \{0, 1, 2, 3\}$
 is a subtring of Z_4 .

2) $S = \{0, \bar{2}, \bar{4}\}$ is subtring of Z_4

$+6$	$\bar{0}$	$\bar{2}$	$\bar{4}$
$\bar{0}$	$\bar{0}$	$\bar{2}$	$\bar{4}$
$\bar{2}$	$\bar{2}$	$\bar{4}$	$\bar{0}$
$\bar{4}$	$\bar{4}$	$\bar{0}$	$\bar{2}$

$\cdot 6$	$\bar{0}$	$\bar{2}$	$\bar{4}$
$\bar{0}$	$\bar{0}$	$\bar{0}$	$\bar{0}$
$\bar{2}$	$\bar{0}$	$\bar{4}$	$\bar{2}$
$\bar{4}$	$\bar{0}$	$\bar{2}$	$\bar{4}$

Since, $a, b \in S, a, b \in S \& a, b \in S$,
 Hence, $S = \{\bar{0}, \bar{2}, \bar{4}\}$ is subtring.

* Lattices:-

POSET:- consider a relation R on set A satisfying the following properties.

i) R is reflexive.

$$\forall x \in A, (x, x) \in R,$$

ii) R is antisymmetric

$$\forall x, y \in A \quad (x, y) \& (y, x) \in R \rightarrow x = y.$$

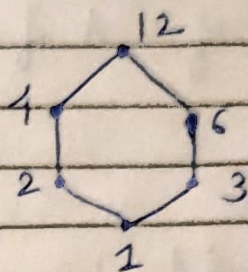
iii) R is transitive.

$$\forall x, y, z \in A \quad \text{if } (x, y) \& (y, z) \in R \Rightarrow (x, z) \in R.$$

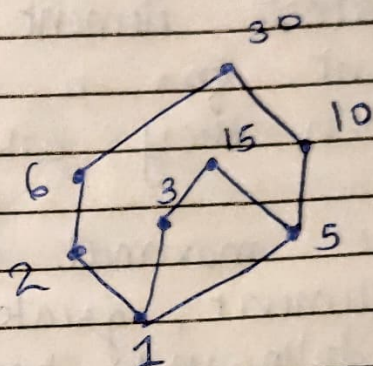
Then R is called a partial order relation & set R together with partial order set or poset.

Draw the Hasse diagram of D_{12} with div. $(D_{12}, |)$

$$D_{12} = \{1, 2, 3, 4, 6, 12\}$$



$$D_{30} = \{1, 2, 3, 5, 6, 10, 15, 30\}$$



1) Comparable element :-

If A is a set & R is partial ordered relation on A then elements $a, b \in A$ are said to be comparable if $a \leq b$ or $b \leq a$. In $(D_{12}, |)$ 1 & 2 are comparable elements but 2 & 3 are non-comparable elements.

2) Maximal element.

An element $a \in A$ is called maximal element of A if there is no $e \in A$,

such that $c \leq a$

4) Greatest element:-

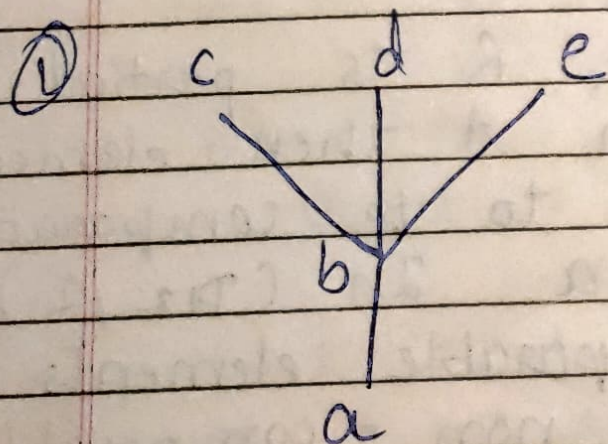
An element $a \in A$, is called greatest element of A if $x \leq a$ $\forall x \in A$.

5) Least element:-

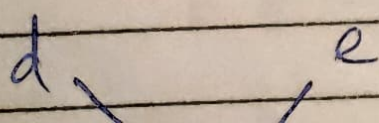
An element $a \in A$ is called least element of A if $a \leq x, \forall x \in A$.

Note:- The greatest element & the least element for a poset if exists then they are unique.

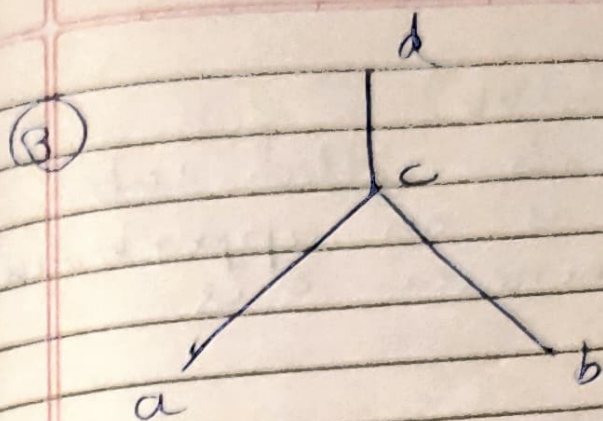
e.g.:- Find the maximal element, minimal element, greatest & least for the following posets.



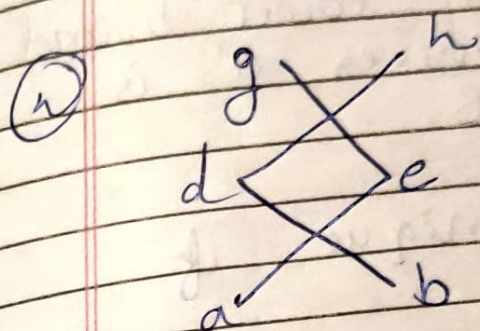
- ① minimal element:- a
- ② maximal element:- c, d, e
- ③ greatest element:- doesn't exist
- ④ least element:- a



- ① minimal:- a, b
- ② maximal:- d, e



- (1) minimal = a
- (2) max = d
- (3) greater = d
- (4) least = a



- minimal = a, b
- max = g, h
- greater = DNE
- least = DNE

Note:- The greatest element is denoted by '1' & least element is denoted by '0'.

Consider a poset A & subset B of A ($B \subseteq A$) then,

- (1) lower bound:- An element $a \in A$ is called lower bound of B if $a \leq b$ for all $b \in B$.
- (2) upper bound:- An element $a \in A$ is called as upper bound of B if $b \leq a$ for all $b \in B$.

3) Least upper bound:-(lub)

An element $a \in A$ is called lub. if lub of B if a is an upper bound of B & $a \leq a'$ whenever a' is upper bound of B .

4) Greatest:- if a is an lower bound of B & $a \geq a'$ whenever a' is a lower bound of B .

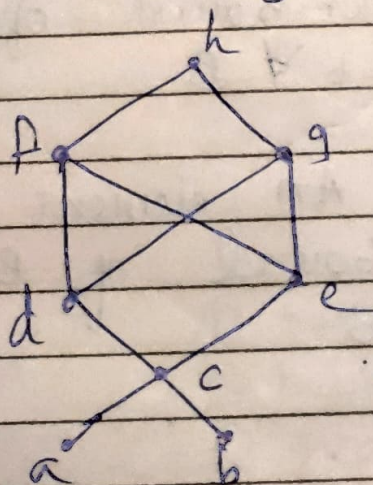
Note:- glb & lub are unique if they exists.

Consider a poset $A = \{a, b, c, d, e, f, g, h\}$ find all upper bound, lower b, lub, glb of A .

1) $B = \{a, b\}$

2) $B = \{c, d, e\}$

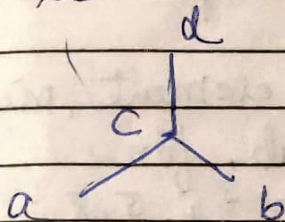
where hasse diag.



1) $B = \{a, b\}$
 $lb = \text{ONE}$
 $glb = \text{ONE}$
 $ub = \{c, d, e, f, g, h\}$
 $lub = e$

2) $B = \{c, d, e\}$
 $lb = \{a, b\}$
 $glb = \text{ONE}$
 $ub = \{f, g, h\}$
 $lub = \text{DNE}$

Q. Give an eg of a poset which has greatest element but not least ele.



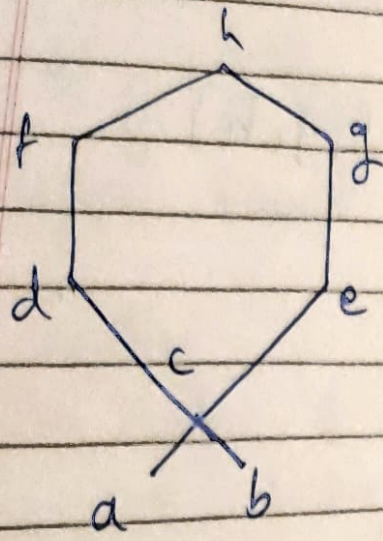
greater = d
 least = DNE.

Greatest ele is d, but as there exist more than one minimal element least ele doesn't exist.

DNE $\rightarrow$ does not exist.

$B_1 = \{d, e\}$

$B_2 = \{f, g, d\}$

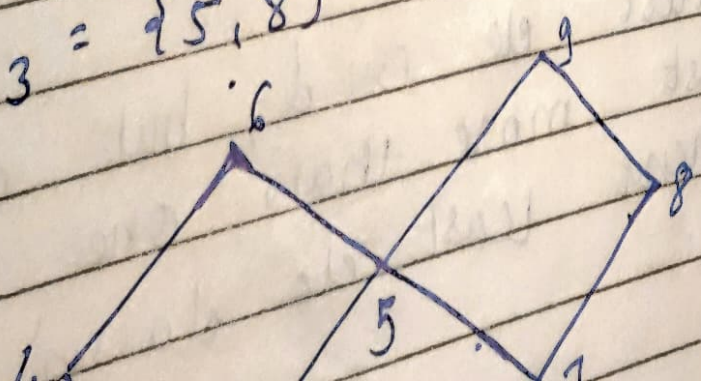


lb, ub, glb, lub.

$\Rightarrow$ minimal element = a, b
maximal elt = h
least elt = DNE
greatest element = h

	$B_1 = \{d, e\}$	$B_2 = \{f, g, d\}$
ub	$\{f, g, h\}$	ub = h
lub	DNE	lub = h
lb	$\{a, b, c\}$	lb = $\{a, b, c\}$
glb	c	glb = c

find least elt, greatest element, min, max elt
Also find lb, ub, glb, lub, for
 $B_1 = \{3, 4, 6\}$
 $B_2 = \{4, 5, 8\}$
 $B_3 = \{5, 8\}$



	$B_1 = \{3, 4, 6\}$	$B_2 = \{4, 5, 7\}$	$B_3 = \{5, 8\}$
ub	6	8 DNE	9
lub	$\{1, 2, 3\}$	DNE	9
lb	6	5 DNE	7
glb	3	DNE	7

* **Join** :- ($\vee$) :-

$$\text{lub } \{a, b\} = a \vee b$$

read as 'a join b. - (a or b)'

* **Meet** ($\wedge$) :-

$$\text{glb } \{a, b\} = a \wedge b$$

(a and b)
read as a meet b.

* **Lattice** :-

Let $(L, \leq)$ is poset. Let (a, b) are any two ele of L , then L is said to be lattice if every pair $\{a, b\}$ has lub and glb.

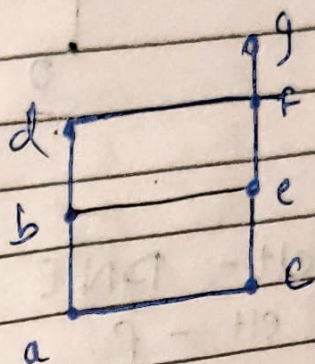
* Determine whether fol. poset is lattice or not.



since, $b \vee e$ does not exist
 $\Rightarrow$ given poset is not a lattice.

H.P. / Q

check whether fol. poset is lattice or not



Types.

* Bounded lattice:-

A lattice L is said to be bounded if it has greatest element (1) and least element (0).

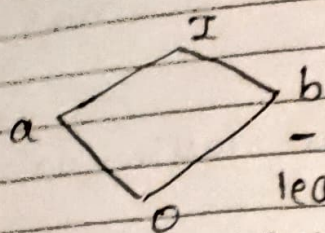
H.P. / $\Rightarrow$

(gub)

$\vee$	a	b	c	d	e	f	g
a	a	b	c	d	e	f	g
b	b	b	d	d	g	f	g
c	.	b	e	d	.	.	.
d	.	.	.	d	.	.	.
e	.	.	.	.	e	.	.
f	.	.	.	.	.	f	.
g	.	.	.	.	.	.	g

Bounded Lattice

1)

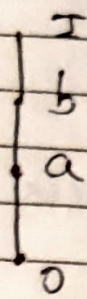


- bounded lat.

least elt = 0

greatest elt = 1

2)

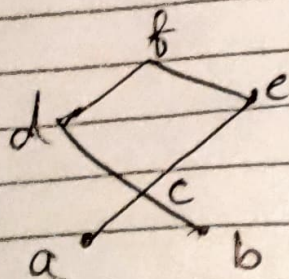


- bounded lat.

- le = 0

- ge = 1

3)



least elt - DNE

greatest elt - f.

∴ NOT bounded lattice.

Sub-lattice.

Let L be a lattice, a non-empty subset S of L is called sub lattice if $a \wedge b, a \vee b, a \wedge b \in S$ whenever $a, b \in S$.

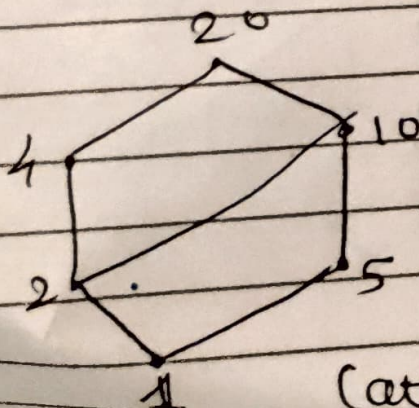
Eg. Let $L = D_{20} = D_{20}$ } $S = \{1, 4, 5, 20\}$

check whether S is sublattice of

L or not,

⇒ $D_{20} = \{1, 2, 4, 5, 10, 20\}$

Hasse Diag



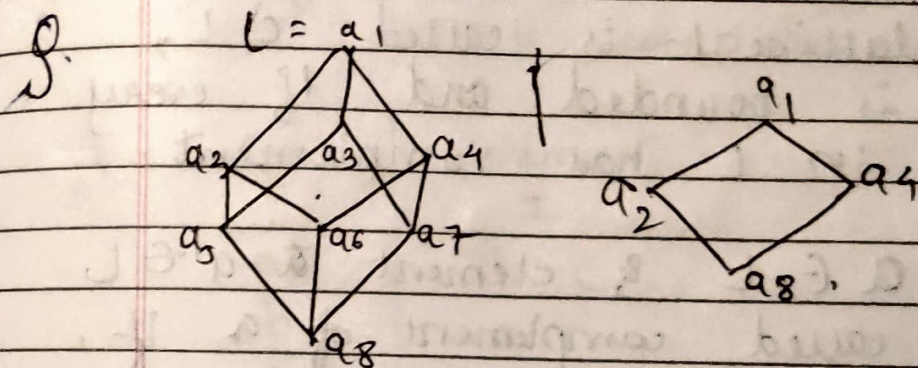
Lattice L .

	V	1	4	5	20
least	1	1	4	5	20
upper	4	4	4	20	20
bound	5	5	20	5	20
	20	20	20	20	20

$\wedge$	1	4	5	20
1	1	1	1	1
4	1	4	1	4
5	1	1	5	5
20	1	4	5	20

Since for $\forall a, b \in S$
 $a \wedge b, a \vee b \in S$

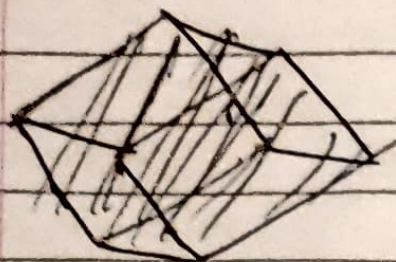
$\Rightarrow S$ is sublattice.



$\Rightarrow L = \{a_1, a_2, a_3, a_4, a_5, a_6, a_7, a_8\}$

$S = \{a_1, a_2, a_4, a_8\}$

$\vee$	a_1	a_2	a_4	a_8
a_1	a_1	a_1	a_1	a_1
a_2	a_1	a_2	a_4	a_2
a_4	a_1	a_1	a_4	a_4
a_8	a_1	a_2	a_4	a_8

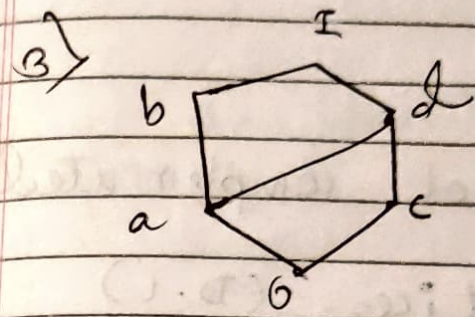
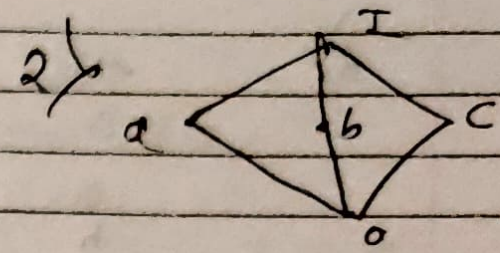
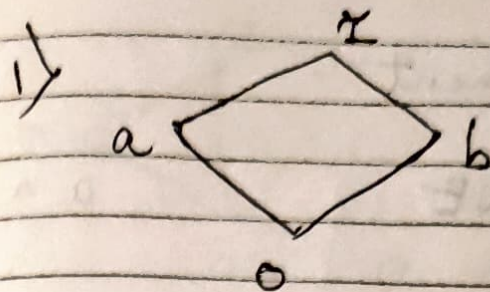


$\wedge$	a_1	a_2	a_4	a_8
a_1	a_1	a_2	a_4	a_8
a_2	a_2	a_2	a_6	a_8
a_4	a_2	a_6	a_4	a_8
a_8	a_8	a_8	a_8	a_8

Since $a_2 \wedge a_4 = a_6$ & $a_6 \notin S$
 $\Rightarrow S$ is not a sub lattice.

Complemented Lattice (C.L)

A lattice L is called C.L, if it is bounded and if every



⇒ 1)

elt	complement
0	I
a	b
b	a
I	0

$$\begin{aligned}
 0 \vee I &= I \\
 0 \wedge I &= 0 \\
 a \vee b &= I \\
 a \wedge b &= 0
 \end{aligned}$$

∴ It is complemented lattice.

elt	element
0	I
a	DNE
b	c
c	b
d	DNE.
I	0

$a \wedge$

Given lattice is not complemented.

Distributive lattice. (D.L)

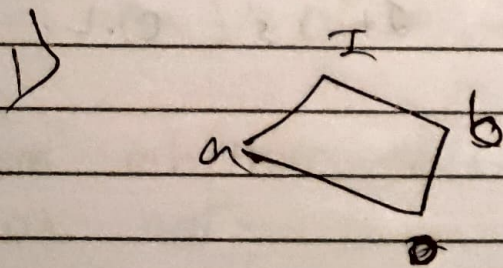
A lattice L is called D.L if for every any element a, b, c of L , it satisfies following D. property

$$(i) \quad a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$$

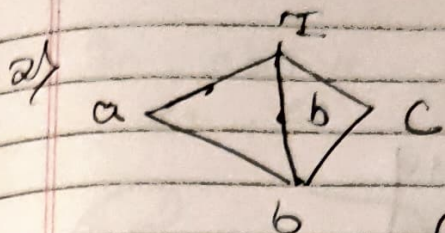
$$(ii) \quad a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$$

Remark: A lattice L is said to be distributive if every element of L has at most one complement.

Check whether it is D. or not,



It is D.L since for any three elements it satisfies Dist. property.



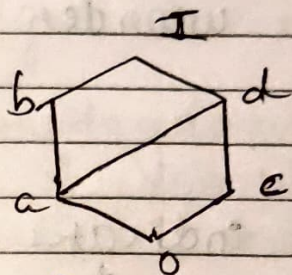
It is not d.L.

$$a \wedge (b \vee c) = a \wedge I = a$$

$$(a \wedge b) \vee (a \wedge c) = 0 \vee 0 = 0$$

$\therefore a \wedge (b \vee c) \neq (a \wedge b) \vee (a \wedge c)$
 $\Rightarrow L$ is not d.L.

3)
HW



a, b, c

b, d

a, b, d

a, c, d

H.W

a, b, c

$$\textcircled{1} a \wedge (b \vee c) = a \wedge I = a$$

$$(a \wedge b) \vee (a \wedge c) = a \vee 0 = a$$

$a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c) \rightarrow L$ is distributive lattice

a, d, c

$$\textcircled{2} a \wedge (d \vee c) = a \wedge d = a$$

$$(a \wedge d) \vee (a \wedge c) = a \vee 0 = a \rightarrow L$$
 is distributive

$\textcircled{3} c, b, d$

$$c \wedge (b \vee d) = c \wedge I = c$$

$$(c \wedge b) \vee (c \wedge d) = 0 \vee c = c$$

$$c \wedge (b \vee d) = (c \wedge b) \vee (c \wedge d)$$

L is distributive lattice.

$\textcircled{4} a, c, d$

$$a \wedge (c \vee d) = a \wedge d = a$$

$$(a \wedge c) \vee (a \wedge d) = 0 \wedge a = a$$

$$a \wedge (c \vee d) = (a \wedge c) \vee (a \wedge d)$$

L is distributive lattice.